Swiggy Sales Analysis Project Report

1. Project Overview

This project focuses on analyzing Swiggy food delivery sales data to derive meaningful business insights using **SQL**, **dimensional modeling**, and **Power BI dashboards**. The objective is to clean raw transactional data, design a scalable **Star Schema**, calculate key performance indicators (KPIs), and visualize trends related to revenue, orders, cities, restaurants, dishes, and customer spending behavior.

The project is designed to demonstrate **end-to-end data analytics skills**, from raw data handling to executive-level reporting.

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2. Business Objectives

- Ensure high data quality through cleaning and validation
- Build a performance-optimized analytical data model
- Track sales performance across time, location, restaurant, and cuisine
- Identify top-performing cities, restaurants, and dishes
- Understand customer spending patterns and ratings

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3. Data Source

- Raw data table: **swiggy**
- Data contains:
    - State, City, Location
    - Order Date
    - Restaurant Name
    - Category (Cuisine)
    - Dish Name
    - Price (INR)
    - Rating & Rating Count

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4. Data Cleaning & Validation (SQL)

Data quality checks were performed before analysis:

4.1 Null Value Check

- Verified missing values in all critical columns such as state, city, order_date, price, rating, etc.

4.2 Blank / Empty Value Check

- Identified records with empty strings that could impact aggregation accuracy

4.3 Duplicate Detection & Removal

- Detected duplicate rows using GROUP BY on all business-critical columns
- Removed duplicates using ROW_NUMBER() while retaining one valid record

This ensured a **clean and reliable dataset** for analysis.

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5. Data Modeling – Star Schema

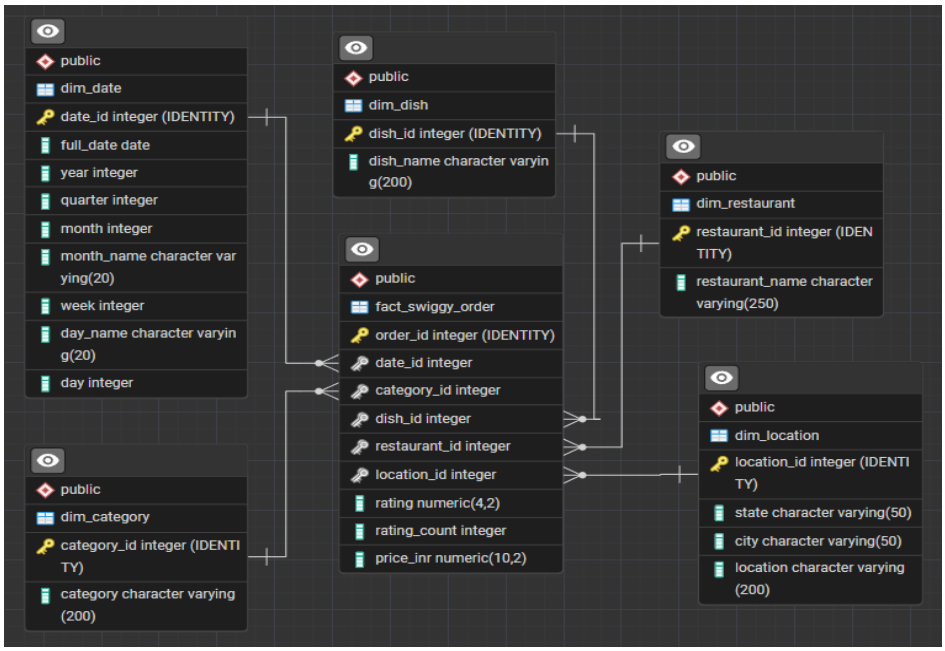
A **Star Schema** was implemented to support efficient BI reporting.

Dimension Tables

- **dim_date** → Date, Year, Quarter, Month, Week, Day Name
- **dim_location** → State, City, Location
- **dim_restaurant** → Restaurant Name
- **dim_category** → Cuisine / Category
- **dim_dish** → Dish Name

Fact Table

- **fact_swiggy_order**
  - Price_INR
  - Rating
  - Rating_Count
  - Foreign keys to all dimensions



This structure improves: - Query performance - Data readability - Scalability for BI tools like Power BI

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6. KPI Development

The following KPIs were calculated using SQL and visualized in Power BI:

Core KPIs

- **Total Revenue:** 53.00M INR
- **Total Orders:** 197K
- **Average Rating:** 4.34
- **Average Order Value (AOV)**

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7. Analytical Insights

7.1 Time-Based Analysis

- Monthly revenue and order trends show seasonal fluctuations

month_name character varying (20)	revenue text	total_order bigint
January	6.82M I...	25393
February	6.27M I...	23292
March	6.57M I...	24400
April	6.59M I...	24584
May	6.79M I...	25188
June	6.51M I...	24383
July	6.65M I...	24936
August	6.79M I...	25227

- Q2 recorded the highest revenue contribution

quarter integer	revenue text	total_order bigint
1	19.66M I...	73085
2	19.90M I...	74155
3	13.44M I...	50163

- Clear weekday vs weekend order behavior patterns

day_name character varying (20)	revenue text	total_order bigint
Monday	7.44M I...	27568
Tuesday	7.36M I...	27413
Wednesday	7.54M I...	28284
Thursday	7.66M I...	28450
Friday	7.58M I...	28285
Saturday	7.78M I...	28933
Sunday	7.64M I...	28470

7.2 Location-Based Analysis

- **Bengaluru** leads in total revenue
- Other high-performing cities include Lucknow, Hyderabad, and Mumbai

city character varying (50)	total_order bigint	total_revenue text
Bengaluru	20072	5.46M INR
Mumbai	10507	3.02M INR
Hyderabad	10309	3.02M INR
Jaipur	10285	2.50M INR
Lucknow	10192	3.12M INR
New Delhi	10191	2.83M INR
Ahmedabad	10175	2.82M INR
Chandigarh	10060	2.80M INR
Kolkata	10044	2.66M INR
Chennai	10042	2.64M INR

- Revenue contribution varies significantly across states

state character varying (50)	total_revenue text
Karnataka	5.46M INR
Uttar Pradesh	3.12M INR
Telangana	3.02M INR
Maharashtra	3.02M INR
Delhi	2.83M INR
Gujarat	2.82M INR
Punjab	2.80M INR
West Bengal	2.66M INR
Tamil Nadu	2.64M INR
Rajasthan	2.50M INR

7.3 Restaurant Performance

- KFC and McDonald's are top revenue-generating restaurants

restaurant_name character varying (250)	total_order bigint	total_revenue numeric
KFC	12957	4245461.78
McDonald's	13528	3342455.58
Pizza Hut	6529	2133265.69
Burger King	7115	1900518.09
Domino's Pizza	5489	1832985.32
Olio - The Wood Fired Pizzeria	3239	1232731.00
LunchBox - Meals and Thalis	4700	1101141.00
Baskin Robbins - Ice Cream Desse...	4197	860591.94
Faasos - Wraps, Rolls & Shawarma	3256	780215.00
The Good Bowl	2665	673343.00

- Certain restaurants have lower order volume but higher AOV

7.4 Dish & Category Analysis

- Top dishes include *Bold BBQ Veggie*, *Korean & Thai Rolls*, and *Paneer Butter Masala*
- Recommended and Main Course categories contribute the highest revenue

7.5 Customer Spending Insights

Customers were segmented into spending ranges: - Under 100 - 100–199 - 200–299 - 300–499 - 500+

Most orders fall within the **200–299 INR** range, indicating a mid-price preference.

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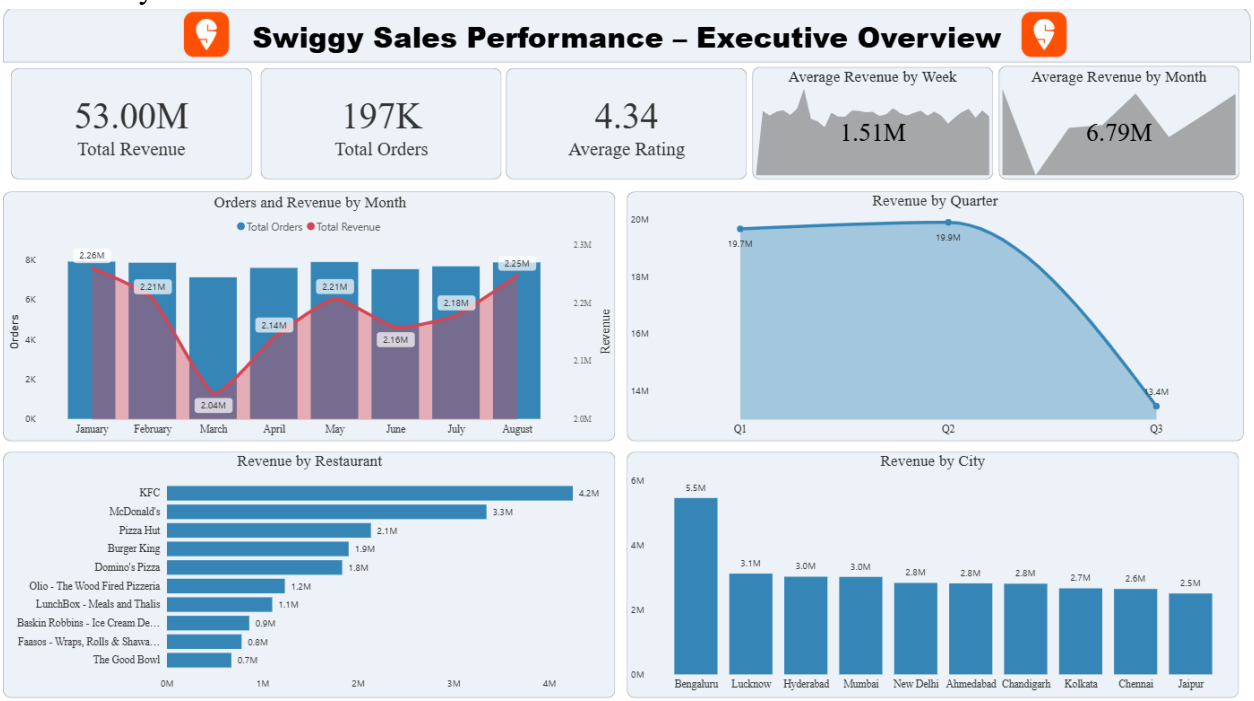
8. Power BI Dashboard Overview

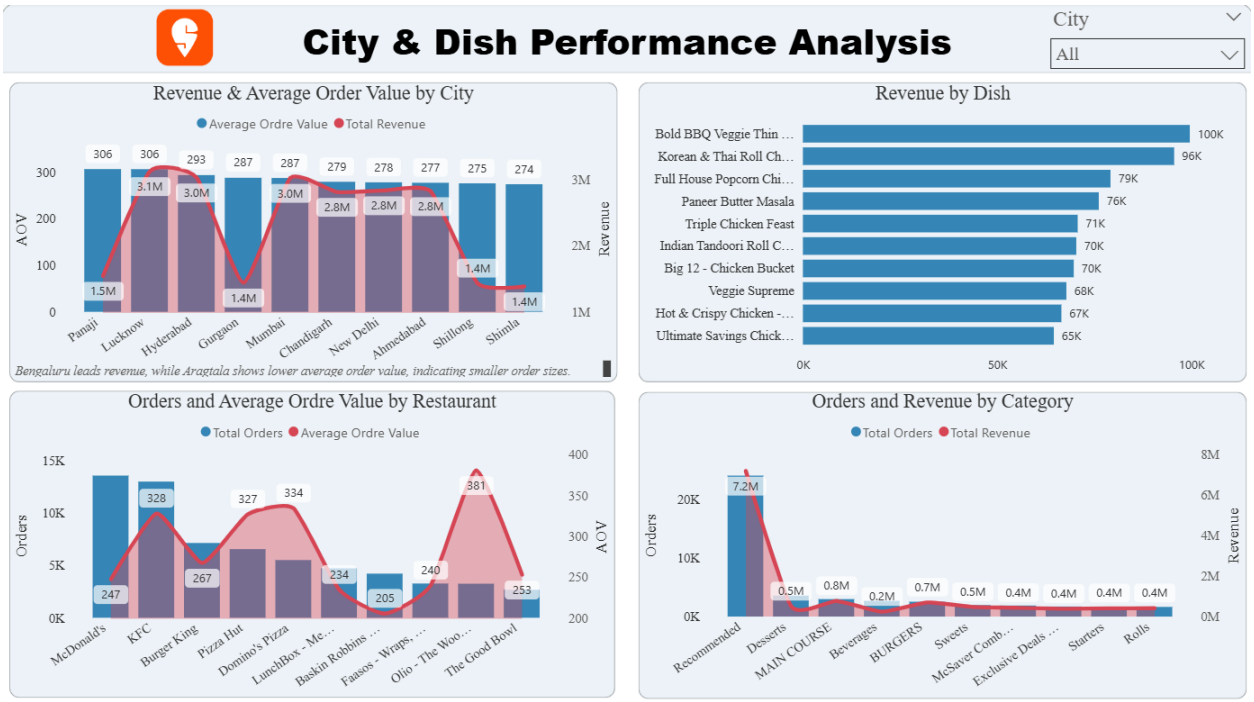
Page 1: Sales Overview Dashboard

- KPI Cards: Total Revenue, Total Orders, Average Rating
- Monthly & Quarterly Revenue Trends
- Revenue by City
- Revenue by Restaurant



Page 2: City & Dish Performance Analysis

- Revenue & AOV by City
- Top Dishes by Revenue
- Orders & AOV by Restaurant
- Orders & Revenue by Category



Interactive slicers were used for **City filtering**, enabling dynamic analysis.

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9. Tools & Technologies Used

- **SQL (PostgreSQL)** – Data cleaning, transformation, modeling, analysis
- **Power BI** – Interactive dashboards and data visualization
- **Star Schema Modeling** – Analytical data design
- **CSV** – Source data ingestion into PostgreSQL

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10. Key Outcomes

- Built a complete analytics pipeline from raw data to dashboards
- Demonstrated strong SQL skills including joins, CTEs, window functions, and aggregations
- Designed professional Power BI dashboards suitable for business stakeholders
- Delivered actionable insights on sales, customers, and product performance

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11. Conclusion

This Swiggy Sales Analysis project showcases real-world data analytics skills required for a **Data Analyst role**, including data cleaning, modeling, KPI creation, and storytelling through dashboards. The solution is scalable, business-ready, and aligned with industry best practices.